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MILITARY SPECIFICATION

MARKER, TRAFFIC LINE, TRUCK MOUNTED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a truck mounted traffic line marker capable of painting a 36 inch wide strip at a speed of 10 miles per hour.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-S-300	Sheeting and Tape, Reflective: Non-exposed Lens
TT-P-85	Paint, Traffic, and Airfield Marking, Solvent Base
TT-P-87	Paint; Traffic, Premixed Reflectorized
UU-T-81	Tags, Shipping and Stock
KKK-T-2108	Trucks and Truck Tractors: Commercial, Diesel and Gasoline Engine Driven, 24,000 to 32,000 GVW, 4X2
PPP-B-636	Box, Shipping, Fiberboard
PPP-T-60	Tape; Packaging, Waterproof
PPP-T-97	Tape; Pressure Sensitive Adhesive, Filament Reinforced

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-B-131	Barrier Material; Water Vaporproof, Flexible, Heat-Sealable

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODSAC, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-H-704	Treatment and Painting of Material
MIL-H-775	Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of
MIL-C-3600	Compressor, Air and Gas (Except Oxygen), Packaging of
MIL-H-3992	Hose and Hose Assembly, Rubber; Air and Vacuum Brake, Automotive
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-E-10062	Engines: Preparation for Shipment and Storage of
MIL-G-10924	Grease, Automotive and Artillery
MIL-H-13531	Hose, Rubber and Hose Assembly, Rubber (Hydraulic Flexible)
MIL-V-13811	Varnish, Waterproofing, Electrical Ignition

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-642	Identification Marking of Combat, and Tactical Transport Vehicles
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods
MIL-STD-1223	Administrative Wheeled Vehicles Treatment, Painting, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards
MS 51317	Light, Warning, Vehicular Rotating, D.C.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York,

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New York 10016.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules

(Application for copies of the above publication should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washington, D.C 20036.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code - Section VIII

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Preproduction model. This specification provides for preproduction model inspection.

3.2 Description. Unless otherwise specified (see 6.2), the marker shall consist essentially of a pavement striping machine, mounted on a gasoline engine driven (GED) truck, capable of applying, at a speed of 10 miles per hour (mph), a 6 inch to 36 inch wide stripe of traffic striping paint and reflective media. Components shall include (a) a 4 x 2 chassis with a gross vehicle weight (GVW) between 24,000 and 32,000 pounds, (b) an air compressor, (c) three 150 gallon capacity paint supply tanks, (d) a 2,000 pound capacity media container and complete media dispensing system to all paint gun stations. Paint cleaner equipment shall be mounted on the platform. An adjustable striping mechanism with separate platform and mounted controls for the striping mechanism with separate platform and mounted controls for the striping mechanism, and intermittent mechanism for the spray guns and media dispensers, inter-communication system, paint filters, air lines, piping and all other accessories necessary to provide a striping machine complete and operable as specified herein shall be furnished.

3.3 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would affect the performance or maintainability of individual components of the overall assembly.

3.4 Protective treatment. When materials must be used in construction that are subject to deterioration from climatic and environmental conditions, materials shall be protected against such deterioration in a manner that will not adversely affect operation of the marker. Protective coatings that will crack, ship, or scale with age or extreme climatic changes shall not be used. Wood treatment and undercoating shall comply with MIL-STD-1223. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting and data plates shall be in accordance with MIL-STD-1223.

3.5 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification to parts or the

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marker.

3.6 Construction. The marker shall be complete so that when it is in operation it will perform all operations for which it was designed. Construction shall be free from any characteristics or defects that will prevent the marker from passing the examination and tests in section 4.

3.7 Design. The marker shall be a truck mounted unit designed to apply paint only, a paint and reflective media mixture, or paint and reflective media separately and simultaneously on a paved surface varying from level to a 15 degree slope. Design shall provide a mechanism for application of reflective media over the painted stripe(s) and to apply the stripe(s) intermittently.

3.8 Safety devices. The marker shall be fitted with safety devices or covering for all parts that present safety hazards. The devices shall include guards for belts, pulleys, chains, gears, and sprockets. All guards shall be designed to provide access to the guarded parts.

3.9 Detachable components. To permit rapid removal, cleaning, and replacement, all detachable components of the fluid and pneumatic systems shall incorporate readily attachable and detachable threaded connectors and fittings.

3.10 Performance. Marker shall paint continuous or intermittent lines as desired when operated in the normal manner. The line shall be of uniform film thickness and width. Sufficient paint or binder shall be deposited (not less than 1 gallon per 100 square feet) to form a continuous line with complete surface coverage in one striping. The marker shall be capable of depositing reflective media at a rate of not less than 10 pounds per 100 square feet and capable of depositing granules at a rate of not less than 1.5 pounds per 100 square feet at a road speed of 10 mph. The marker shall be capable of applying traffic marking paints of all types, both reflective and non-reflective, in a line 6 inches to 36 inches wide in increments of 6 inches. It shall be capable of marking a solid line or a skip line, or a combination of solid and skip lines, of any width within its range. It shall be capable of marking 2 colors at the same time with the above combinations and as specified in TT-P-87, using yellow and white colors for street marking. The marker shall also be capable of marking runways with TT-P-85. The marker shall have variable marking speeds of 2 to 10 mph.

3.11 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.12 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.13 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of the marker.

3.14 Adjustment and repairs. Marker shall be constructed so that adjustments

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and repairs can be easily made with standard tools by operating personnel.

3.15 Chassis. The truck chassis shall be in accordance with KKK-T-2108, Type I, minimum 24,000 pounds GVW except as specified herein. The following items shall be furnished in addition to the basic requirements of KKK-T-2108.

a. Heavy duty cooling system.

b. An alternator of sufficient current capacity to operate all of the electrically operated components on the marker shall be furnished.

c. Tilt cab.

d. Power steering.

e. Automatic transmission.

f. Two-speed rear axle.

3.16 Platform. The platform shall have a minimum length of 16 feet and a width of 94 ± 2 inches and shall have a floor plate of not less than 11 gauge, safety tread, steel, supported by structural steel channel sills and formed steel "U" shaped crossbars. The platform shall extend at least to the end of the truck frame. The wheels shall be entirely covered by the platform. A safety tread equipped ladder with rails shall be mounted on the rear of the platform. A welded hand rail approximately 3 feet high shall be installed around the edges of the platform with an entrance at the head of the ladder.

3.17 Air compressor assembly. A skid mounted engine driven air compressor capable of running all of the marking equipment and accessories simultaneously shall be provided. Required supply lines, air receivers and other necessary compressor accessories shall be included. The entire unit shall be enclosed in a weather resistant housing with folding or flat sliding doors, complete with spring loaded locking handles. The unit shall be mounted on vibration isolators.

3.18 Compressor engine. Unless otherwise specified (see 6.2), the compressor shall be driven by an internal-combustion gasoline engine. The engine shall be equipped with generator or alternator, battery, self-starter, and a spring loaded safety coupling between the engine and compressor. The compressor engine, when delivering no more than 80% of the engine manufacturer's rated continuous horse-power (HP) at the engine manufacturer's rated speed, shall be capable of driving the compressor under the most severe operating conditions.

3.19 Compressor. The air compressor shall be two stage rotary type, capable of providing not less than 125 standard cubic feet of air per minute, at 100 pounds per square inch (psi) gauge pressure. An adjustable automatic air unloader shall be provided to allow the engine to idle when the compressor is not compressing air.

3.20 Air receiver. A steel pulsation chamber, ASME rated, shall be provided. This chamber shall act as an air receiver and be fabricated to operate on air pressures up to 150 psi. The chamber shall be fitted with proper size inlet and outlet ports and shall have attached a 160 psi pressure gauge with a 230° minimum length of arc for a scale range of 0-160 and a 110 psi safety valve shall be furnished. The outlets shall be provided with a 125 psi nonferrous

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plug or 90 degree valve, located conveniently to the operator. A condensate bleeder valve shall be installed in the bottom of the tank.

3.21 Paint tanks. The marker shall include three 150 gallon capacity tanks complying with the ASME Code. Tanks shall be fabricated to withstand an internal working pressure of 110 psi. The tanks shall be securely mounted on the platform, each equipped with all necessary gauges and valves. The tanks shall be constructed so as to provide easy access for cleaning and repairing. A separate opening of at least 2-1/2 inches in diameter shall be provided. The three tank covers shall be completely interchangeable. A motor of at least 1/2 horse power to drive agitators shall be provided on each tank. The agitator paddles shall be approximately the full diameter of the tank opening.

3.22 Reflective media tank. A steel tank(s) with a capacity of 2,000 pounds of reflective glass spheres shall be provided. An opening shall be provided at the bottom of the tank of sufficient size to supply media to each dispenser on the center line assembly. The media tank shall be mounted at the most convenient point on the truck bed for loading and to allow media to be dispensed through flexible conductors directly to the media dispensers. The tanks shall have a minimum rating of 110 psi and comply with ASME code.

3.23 Paint spray guns. Paint spray guns shall be heavy duty, bleeder type, air controlled or manufacturer's standard system with valving solenoids mounted near the guns. Assembly of guns shall be capable of painting a stripe 6 inches to 36 inches wide at .015 inch dry thickness at 10 mph. Each gun shall be adjustable vertically and laterally for line width and spacing. The spray equipment shall be electrically controlled by means of switches and solenoid valves. Media dispensers shall be controlled by the same timing device for synchronized gun and dispenser operation.

3.24 Media dispensers. Media dispensers shall be capable of performing as herein required. The pattern for media dispensing shall be 6 inches wide to 36 inches wide. The dispensers shall provide a constant flow and even distribution of media under all striping conditions as specified. A molded rubber or metal shroud(s) as required shall be provided to insure line definitions. The media volume control shall be readily adjustable, without interchanging of parts, to supply infinite variations in volume of media as specified, up to at least 10 pounds per 100 square feet of spheres or 1.54 pounds of granules as specified. The dispenser(s) shall be capable of distributing media uniformly over the entire length and width of stripe being painted, with no skipping of painted areas, no piling of media, and no media in areas not painted as required.

3.25 Outrigger assembly. The outrigger assembly shall be equipped with at least 6 paint spray guns and 6 media dispensers. The carriage and striping mechanism shall be mounted on a rigid steel telescopic assembly mounted underneath the truck bed. The outriggers shall be capable of extending away from the left edge of the platform for a minimum distance of 2 feet. The outrigger shall be adjustable and provided with means for locking it in position. Operations required by seated operator shall provide for full visibility ahead of the truck and full visibility of the operations of the guns and media dispenser(s). Means shall be provided to prevent outrigger movement due to road shock. The outrigger assembly shall be capable of being raised a minimum of 12 inches from ground level by means of air or hydraulic cylinder(s) and be retractable parallel to the truck platform for transporting.

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3.26 Cleaner equipment. A cleaner tank complying with ASME Code, with all necessary connections and valves to permit the injection of cleaning fluid through any one gun or all guns, shall be provided. The cleaner tank shall have a capacity of at least 18 gallons. It shall have ample opening for easy filling and shall be equipped with a dusting gun with not less than 6 feet of hose.

3.27 Paint and binder pump. A self-priming pump shall be provided for pumping paints and binder as specified in TT-P-85 and TT-P-87. The pump shall be connected to a single power take off unit or manufacturer's standard system. A connection to each tank shall be provided to permit the filling of the tanks from the bottom. A shut off valve shall be furnished and installed at the pump intake. The pump shall be capable of pumping 35 gallons per minute at 70 Krebs Units viscosity. The connecting lines and valving shall be connected to allow independent flushing after each use.

3.28 Control panel. A control panel convenient to the operator shall be furnished complete with all necessary switches, regulators, gauges, hose fittings. The operation of spraying equipment and related equipment shall be controlled from this panel by means of switches operating electric air valves.

3.29 Intermittent line control. An automatic intermittent line control shall be furnished. The line control of striping operation shall include multiple gun, multiple color and media dispensing units. The timer mechanism shall be driven by means of a ground contact wheel or manufacturer's standard method. The timer assembly shall include a variable drive, gear reduction and electric clutch to transmit the drive to the camshaft. Provisions shall be made to speed up the rotation speed of the camshaft for short intervals of time to advance the paint cycle. Means shall also be provided to retard the paint cycle. The camshaft shall have means to control the required striping pattern. The controlling device shall actuate switches which in turn control the guns and media dispenser(s) through solenoid operated air valves. Precision control of starting and stopping of paint and media shall be provided. A return to zero mechanism shall be incorporated on the camshaft which returns to a starting position automatically when the drive clutch is disengaged. This operation shall have a locking device to positively hold in the zero position. The timer shall be located within the under side of the truck chassis and controlled by means of a control center adjacent to the operation. The control center shall include a master switch and independent gun switches as required. Advance and retard switches shall be located on the control panel. The master switch shall turn all guns and media dispensers on or off at any time and allow the operator to pre-set the gun and dispenser switches as required. The gun and dispenser switches shall be capable of being selected for either solid or intermittent line control as required for street or runway marking.

3.30 Operator's seat. A seat with back rest shall be provided for the operator. The seat shall be located on the platform in a position for convenient operation of the control panel and for ease of operation of the striping operation.

3.31 Intercom system. A means of vocal communication shall be provided between the driver of the truck and the operator of striping equipment. The system shall be the two station type (talk-listen). The master station shall be installed in the truck cab and a weather proof station unit installed at the control panel of the platform. A spring return talk-listen foot switch

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shall be installed for operation by the seated striper operator in order for his hands to be free at all times. This system shall be operable from the truck battery and shall be of heavy duty construction with shielded harness which is radio interference free and vibration proof.

3.32 Pointer guide. An adjustable, detachable and collapsible wheel mounted pointer guide, which shall be hinged and so arranged that it can collapse into a convenient transportable position, shall be mounted ahead of the front bumper. The guide wheel shall be equipped with pneumatic rubber tire, and roller or ball bearings. The guide shall be visible to the driver in all positions of use. One flag socket shall be mounted on the front of the pointer guide.

3.33 Auxiliary hand paint spray gun. A gun-type sprayer for hand operation with 30 feet of air hose and 30 feet of paint hose shall be provided and capable of spraying paints as specified. Provisions shall be made to coil and clip hoses to marker when not in use.

3.34 Paint heater. A paint heater shall be provided which shall be capable of heating two colors of paint simultaneously to permit application of the paint as specified. The system shall include manufacturer's standard heating system or heat transfer device, thermostats, temperature indicators, cleaning equipment, hot water heated hoses and controls. The heat transfer device shall consist of a shell and tube heat exchanger. The heat exchanger shall be piped in a counterflow single pass arrangement for maximum heat transfer between the paint and heating fluid. Paint will only be heated as required to supply the spray guns. A thermostat and solenoid operated valve shall be included to control the paint temperature in the heat exchanger. Suitable adapters and insulated hoses shall be supplied to connect the heater and controls to either the vehicle engine or water cooled air compressor engine. A thermostat shall be provided for the cooling system of the heat generator. The heater shall have the necessary piping running to the cleaner system in such a manner as to independently flush and clean the heat exchanger and hose lines. Provisions shall be made on the control panel to permit switching the entire unit on or off and shall also include two indicator lights; one to indicate when the unit is turned on and the other to indicate when paint is at spraying temperature. The heater shall be capable of raising paint temperature to not less than 160 degrees F in not more than 10 minutes and maintain temperature at not less than 160 degrees F.

3.35 Storage and tool box. A tool box shall be provided and shall be made of metal not less than 14 gage (0.0747 inch) thickness. It shall be complete with a hinged lid and locking device. The tool box shall have storage capacities to include tools, spare parts and hand spray guns. The tool box shall be securely fastened to the marker.

3.36 Flexible hose. Hose for hydraulic lines shall conform to MIL-H-13531, Grade A. Hose for pneumatic lines shall be in accordance with MIL-H-3992. Hose may be nylon, rubber or other materials normally used by manufacturers of marker equipment.

3.37 Lubrication. All bearings (except sealed-for-life type), mating gears, revolving shafts, and sliding parts shall be provided with means for lubrication. The reservoirs of splash type systems shall be fitted with oil level sight gages. Manually operated systems for remote area shall have control handles mounted in an accessible location. All oil holes, grease fittings, and filler caps shall be easily accessible.

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3.38 Electromagnetic Interference. The vehicle shall comply with the requirements of MIL-STD-461, test method RE05.

3.39 Painting. The marker shall be painted yellow, color 13538, conforming to FED-STD-595.

3.40 Instruction plates. Marker shall have permanently affixed instruction plates, as required, describing procedure to be followed in operating or servicing the marker. Plate must be clearly visible to operator while marker is operating. Material shall be the same as identification plate. All words shall be in the English language with plain bold face letters.

3.41 Repair parts and maintenance tools. Such repair parts and maintenance tools as specified (see 6.2) shall be furnished.

3.42 Identification marking. As specified by the procuring activity for the appropriate service (see 6.2), marking and registration numbers shall be in accordance with MIL-STD-642 and/or L-S-300.

3.43 Air transportable. The marker shall be designed so the vehicle with cab and mounted body shall be compatible with the C-130 aircraft for air transportability and shall include being designed in accordance with MIL-A-8421 air transportability requirements for items to be air transported in the C-130 aircraft.

3.44 Emergency amber light. An emergency warning light conforming to MS 51317 shall be provided and mounted with reinforcement on the center line of the cab header section. It shall be operated by a switch located on the instrument panel of the vehicle and have 360 degrees visibility.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Preproduction inspection.

b. Quality conformance inspection.

4.2.1 Preproduction inspection. Preproduction inspection shall be applied to the preproduction sample. Failure of the preproduction sample to pass the examination or one or more preproduction tests shall be cause for rejection. Preproduction inspection shall consist of the examination in 4.3 and all tests in 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being prepared for delivery under the terms of the contract or purchase order. Failure of the machine to pass the examination or one or more tests shall be cause for rejection. Quality conformance inspection shall consist of the following:

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a. Examination (4.3).

b. Inspection of preservation, packaging, packing, and marking for shipment and storage (4.5).

4.3 Examination. The marker and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests.

4.4.1 Operational test. The marker shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and other accessories shall be inspected during the trial period. Paint container shall be hydrostatically tested at two times the working pressure as specified.

4.4.2 Performance test. The marker shall be made ready for normal operation. The marker engine shall be started and allowed to run idle for 5 minutes. The marker shall then be operated over a course including an ascending and descending 15 degree slope, right and left curves. Beginnings, endings, and side boundaries of straight, curved, and intermittent lines shall be checked to determine conformance to 3.10. Control of the marker shall be observed to determine conformance to 3.25 as applicable. Width of stripe shall be measured to determine conformance to 3.10. When furnished, the operation of the media dispenser, pointer guide, intermittent line mechanism, and hand spray gun shall be checked to determine conformance to 3.24, 3.29, 3.32, and 3.33. The overall performance of the marker shall be observed for conformance to this specification. The marker shall apply four continuous stripes and two intermittent stripes following prescribed guide lines at various speeds between 3 and 10 mph, inclusive. Each stripe shall be 1,000 feet long. Paint (as specified in 3.10) shall be used for four of the continuous stripes. Paint of two colors applied simultaneously and reflective media as specified in 3.10 shall be used for the intermittent stripes, which shall include periodic starts and stops of striping and flushing guns with solvent. The striping path shall be varied periodically to prove versatility of the path controlling device. A speedometer shall be connected to the marker for the test. Reflective media shall be uniformly distributed on the paint and shall conform to the width of each stripe. There shall be no excess media which can be brushed off the paint after it dries, and no media deposited on unpainted surfaces. Paint heater shall be observed to determine conformance to 3.34.

4.4.3 Flushing and agitation test. The marker shall be subjected to this test after stripes of paint and reflective media have been applied as specified in 4.4.2. The paint hose and spray head shall be flushed with paint solvent by operating the cleaning assembly. When flushing is concluded, agitator mechanism shall be agitated as required.

4.4.3.1 Effectiveness of agitation. The paint in the tank shall show no evidence of pigment separation or scum formation.

4.4.4 Hand spray paint test. The spray gun shall be connected to the marker and operated by a qualified operator. Paint shall be sprayed over a test piece having an area of at least 50 square feet. The marker shall have the following performance characteristics:

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a. Paint spray. The spray gun shall emit a steady, uniform mixture of paint and air to produce a uniform coating on the test piece, free from runs, skips, and splotches.

b. Spray gun control. Control for the start and stop of the spray shall be immediate and positive acting.

4.4.5 Electromagnetic interference. The marker shall be tested as a complete unit for electromagnetic interference suppression in accordance with MIL-STD-461 Test Method RE05.

4.5 Servicing and adjusting. Prior to acceptance of the vehicle by the Government Inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: Focusing of lights; adjustment of the engine(s), electrical and brake system; filling and charging of battery(s); alignment of front wheels; inflation of all tires; complete lubrication of chassis, engine(s), running gear, all painting equipment including air compressor with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

4.6 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage, loss or breakage in transit. Bolts, nuts, screws, pins, washers and other attaching parts removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on shipping tags conforming to UU-T-81, Type A. The printed tags shall be waterproofed in accordance with MIL-STD-129.

5.1.2 Cleaning and drying. Prior to the application of preservatives or paint, the surfaces involved shall be cleaned in accordance with process C-1 (any applicable process) of MIL-P-116 and dried by the most applicable procedures specified therein. Processes and procedures used shall accomplish thorough cleaning and drying without damage to any part or its preservation.

5.1.3 Preservatives and materials. Preservatives and materials shall be as specified herein or in applicable specifications, standards, etc. References to "P-type" preservatives are to those listed in MIL-P-116. Materials shall be free from defects affecting service ability.

5.1.4 Unpainted surfaces. Unpainted surfaces of such items as bolts, nuts, guide bar and other unpainted exterior ferrous metal surfaces shall be coated with type P-1 preservative. This preservative shall not be used in any appli-

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cation where removal is required. Exposed, unpainted, machined-metal surfaces and threaded surfaces which require occasional turning in normal operation of the unit shall be coated with grease conforming to MIL-G-10924.

5.1.5 Hose and hose fittings. Hose and hose fittings shall be preserved and unit packaged in conformance with MIL-H-775.

5.1.6 Gasoline engine. Gasoline engine shall be preserved in conformance with the applicable requirements of MIL-E-10062 except that the spraying of the engine and accessories with insulation compound is waivered and shall not be performed.

5.1.7 Fuel tank. The fuel tank interior shall be coated with type P-10, Grade 2 preservative oil by either the fogging or spraying methods specified in MIL-P-116. The tank shall be allowed to stand with drain plug open until oil flow ceases. The drain plug shall be reinstalled and the filler cap replaced.

5.1.8 Air compressor. The air compressor and components shall be preserved in accordance with the requirements of MIL-C-3600.

5.1.9 Outrigger. Unpainted exterior ferrous metal surfaces of the outrigger shall be coated with type P-1 preservative. Hose connection openings shall be sealed with pressure-sensitive waterproof adhesive tape conforming to PPP-T-60, Type III, Class 1. Wheel bearings shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.10 Pointer guide. Wheel bearings shall be preserved as specified in paragraph 5.1.9. Disassembly shall be to the extent necessary to reduce cube when packed with the other component parts. Mounting bolts shall be secured to the guide to prevent their loss.

5.1.11 Material tanks. Material tanks shall be cleaned and dried as specified in 5.1.2. All threaded openings shall be sealed with threaded metal or plastic caps or plugs. All non-threaded openings shall be sealed with tape conforming to PPP-T-60, Type III, Class I.

5.1.12 Spray guns. The spray guns shall be preserved in accordance with Method IA of MIL-P-116.

5.1.13 Electrical equipment.

5.1.13.1 Wiring, junction boxes and similar items. Wiring, junction boxes and similar items shall be coated with electrical ignition waterproofing varnish conforming to MIL-V-13811. Exposed ends of wires and terminals, openings into sockets, coupling plugs, and switches shall be sealed with waterproof pressure-sensitive tape conforming to PPP-T-60, Type II, Class I, before the varnish is applied.

5.1.13.2 Electric cables. Electric cables shall be coiled in the minimum safe dimensions and secured in not less than three places with tape conforming to PPP-T-97. The ends shall be wrapped with barrier material conforming to MIL-B-131 secured in place with pressure-sensitive tape conforming to PPP-T-60.

5.1.13.3 Instrument panels. The glass fronts of instruments, instrument panels and dial glasses shall be protected by covering with greaseproof barrier material conforming to MIL-B-121, Grade A and bridging the glasses with pieces

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of fitted plywood or other similar protective material secured in place with pressure-sensitive tape conforming to PPP-T-60.

5.1.14 Tool box. When tools are furnished, they shall be placed in the tool box. The lid of the tool box shall be closed and secured.

5.1.15 Intermediate packaging. Removed parts and technical publications shall be packed in a minimum number of close-fitting fiber boxes conforming to PPP-B-636, type and class optional. The boxes shall be closed in accordance with the Appendix to the box specification.

5.2 Level C. The markers shall be preserved and packaged in accordance with the manufacturer's standard commercial practice.

5.3 Packing.

5.3.1 Level A. The marker, packaged and unpackaged parts removed by disassembly, shall be packed in wood sheathed crates conforming to Type I, Class I, Style A or B of MIL-C-104. Anchoring, blocking and bracing shall be in accordance with MIL-STD-1186 and the appendix to MIL-C-104. All attachments or fixtures which do not facilitate packing shall be removed and packed in the shipping container with the remainder of the equipment.

5.3.2 Level C. The marker, component parts, and tools comprising a complete unit shall be packed to insure carrier acceptance and safe delivery to destination in compliance with Official Classification Committee, Uniform Freight Classification Rules or American Trucking Association, National Motor Freight Classification Rules or with other carrier rules and regulations applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required by the contract or purchase order, marking for identification and shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The marker covered by this specification is intended for use in marking aircraft runways with traffic paint and reflective glass beads.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Treatment and painting requirements (see 3.4).

c. Repair parts and maintenance tools to be furnished (see 3.41).

d. Marking requirements (see 3.42).

e. Required levels of preservation, packaging and packing (see section 5).

f. When diesel engine is required (see 3.2).

g. When diesel engine is required (see 3.18).

h. When power plant heaters are required.

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6.3 Preproduction model. The preproduction model may be a preproduction item or a production item which meets the requirements of this specfication. In either case, the model shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a preproduction model shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

Custodian:

Air Force - 99

Preparing Activity:

Air Force - 99

Project Number:

3825-F152

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-83925A (USAF) Marker, Traffic Line, Truck Mounted

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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